Auction Design and Market Concentration in MakerDAO Liquidations:

Evidence from Black Thursday, the LUNA Crash, and the Tariff Shock

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Abstract

We study keeper market concentration across two liquidation mechanisms in MakerDAO: the English auction (Flip, 2020–2021) and the Dutch auction (Clipper, 2021–2026). Using on-chain data from Ethereum event logs and transaction traces, we document that the transition to Clipper did not reduce market concentration — it substantially increased it under extreme stress conditions. We analyze three crisis episodes: Black Thursday (March 2020), the LUNA Crash (May–June 2022), and the Tariff Shock (April 2025). The Herfindahl-Hirschman Index (HHI) varies dramatically across events: ~1,562 on Black Thursday (cumulative count; hourly weighted HHI: 4,496, Table 4), ~26 on the LUNA Crash, and ~8,371–9,546 on the Tariff Shock. This variation reveals that concentration is not a stable property of the mechanism, but rather a function of the speed and intensity of the shock. The LUNA Crash — the only event with genuine access competition, with 399 keepers and count HHI ~26 — demonstrates that Clipper can function as designed under moderate, prolonged stress conditions. However, under high-magnitude instantaneous shocks such as the Tariff Shock, a single keeper (0xc721) captured 97.68% of the ETH liquidated within a 67-minute window. The speed-concentration design trade-off does not reside in the Flip→Clipper transition, but in the interaction between the mechanism and shock speed: any Dutch auction fast enough to liquidate in seconds will disproportionately reward the actor with the most advanced execution infrastructure. This result is a derivable consequence of the design. Mann-Whitney tests (Monte Carlo, 50,000 permutations) confirm statistically significant differences in the two pairs with adequate statistical power: BT vs. LUNA ($p < 0.001$, $|r| = 0.846$) and LUNA vs. Tariff ($p = 0.005$, $|r| = 0.438$). The BT vs. Tariff Shock comparison (n = 4 hours) is reported as descriptive statistics due to insufficient power (§7.2).

Keywords: MakerDAO, DeFi liquidations, keeper market structure, Dutch auction, Flip mechanism, English auction, Liquidations 1.0, MEV, market concentration, HHI, Black Thursday, LUNA Crash, Tariff Shock

0. Position in the Series

This document is part of a four-paper series whose unifying thread is Lemma 3 of Villega (2026a): for any protocol layer where participation requires a fixed cost K, the number of viable agents n* is decreasing in K. The table below situates each paper in the series. The bold row corresponds to the present document.

Paper	Short Title	Layer	K as	Main Method	Dune Queries	Central Finding	Connects To
P1	**Programmatic deposits and monetary transmission**	**Deposits (Pot/DSR)**	**c_gas + c_audit + κ_coord; K*≈2.93M DAI**	**S,s model; calibration; OLS Newey-West**	**PIT_OU_spread_calibration_v1 (query 7384994)**	**λ≈0.86; δ=0.5703; n*=2 (sDAI+Spark, 86% stock); governance as monetary reaction function**	**P3 (GSM as manipulable parameter); P4 (κ_coord estimated)**

P2	**Auction design and concentration in liquidations**	**Liquidations (Clipper/Flip)**	**MEV infrastructure (latency, gas, bots)**	**HHI; bootstrap; Mann-Whitney Monte Carlo**	**ethereum.logs, ethereum.transactions, ethereum.traces**	**Concentration = f(shock speed); Tariff Shock: $n^*\approx 1$ (0xc721, 97.68% in 67 min); LUNA: $n^*\approx 399$**	**P1 (Lemma 3 instantiated); P3 (analogous infrastructure asymmetry in governance)**
P3	Governance dynamics and information asymmetry	Governance (DSChief/GSM)	Coordination cost of outgoing spell	31 DuneSQL queries; case analysis	p3_01–p3_31; dashboard: dune.com/facundovillega/governance-dynamics	GSM delay as endogenous variable (oscillation 16h–30h, 2024); pre-positioning 10 days before cast; on-chain access asymmetry	P1 (GSM as fixed parameter); P2 (same infrastructure asymmetry pattern); P4 (gsm_delay_hours as κ_coord regressor)
P4	κ_coord as a market variable	Governance (κ_coord empirical)	κ_coord_proxy = mkr_mobilized × gsm_delay_hours	OLS HC3; N=217 spells, 2019–2025	p4_01 (7503629), p4_02 (7486415), p4_03 (7486567)	β(log MKR)=1.13; β(gsm_delay_hours)=0.053; HHI not significant; R^2=0.672	P1 (κ_coord closes K* calibration); P3 (gsm_delay as independent variable)

1. Introduction

Liquidation mechanisms are a critical component of collateralized lending protocols. When collateral value falls below required thresholds, automated systems must efficiently recover protocol debt — ideally without destabilizing underlying asset markets. The design of these mechanisms determines who can participate as a liquidator (keeper), how competitive the market is, and whether the protocol recovers debt at fair prices. These questions are not merely technical: a liquidation market dominated by a small number of actors concentrates systemic risk, creates potential for collusion, and may produce worse outcomes for borrowers and the protocol during stress events.

MakerDAO, one of the largest collateralized debt protocols on Ethereum, has operated two distinct liquidation mechanisms throughout its history. The original Flip auction (Liquidations 1.0) used an ascending English auction format where keepers competed by bidding increasing amounts of DAI — a design requiring upfront capital that favored well-funded participants. Following the Black Thursday failures (March 12–13, 2020), where network congestion allowed a small group of keepers to acquire collateral worth $8.32M at zero cost, MakerDAO introduced the Clipper mechanism (Liquidations 2.0): a descending-price Dutch auction explicitly designed to eliminate capital barriers through flash loan composability, thereby creating a more competitive and distributed liquidation market.

The stated objective of Liquidations 2.0 was to democratize keeper participation. This paper asks whether it succeeded — and offers a nuanced answer. Analyzing three crisis episodes with structurally distinct characteristics, we document that market concentration under Clipper is not a constant but a function of shock speed and intensity. The May–June 2022 LUNA Crash produced genuine competition: 399 keepers participated with count HHI ~26, close to perfect competition. The April 2025 Tariff Shock produced the opposite: a de facto monopoly with HHI ~8,371–9,546 and a single keeper capturing 97.68% of the liquidated ETH. This result is consistent with Lemma 3 of Villega (2026a): endogenous concentration is a structural consequence of any market with heterogeneous fixed costs and asymmetric participation. The empirical estimation of the coordination component κ_coord linking the liquidation layer to governance is developed in Villega (2026d).

This variation between events reveals the central design limitation of Liquidations 2.0: it did not eliminate the conditions for concentration, but shifted them to a dimension — execution speed infrastructure — that is harder to address through mechanism design and only manifests under instantaneous shocks. The result is a market that appears competitive under normal conditions but collapses into monopoly under extreme stress.

The theoretical framework organizing this paper is the application of Lemma 3 of Villega (2026a) to the keeper market. Lemma 3 states that the maximum number of viable agents n* in a market with fixed integration cost K and asymmetric participation is decreasing in K: $n^* \leq D\_total \cdot r / (\rho \cdot K)$. In the liquidation layer, K corresponds to MEV infrastructure costs — execution latency, contract optimization, gas management — and n* is the number of keepers capable of competing within a liquidation window of duration T. Under this interpretation, the Tariff Shock is not an anomaly: it is the case where T is sufficiently short for K to become binding and only $n^* \approx 1$ keeper is viable. The LUNA Crash is the opposite case: a long T implies K is a low effective barrier, and $n^* \to$ open competition. This reading connects the empirical results of §§5–6 with the formal framework of Villega (2026a) and with the κ_coord estimation in Villega (2026d).

2. Institutional Framework

2.1 The Flip Mechanism (Liquidations 1.0)

The Flip auction operated as a two-phase English auction. In the tend phase, keepers bid increasing amounts of DAI for the full collateral lot. In the dent phase, keepers bid to accept decreasing amounts of collateral for the fixed DAI amount. This mechanism required keepers to hold DAI upfront, creating a capital barrier to participation.

2.2 Black Thursday (March 12–13, 2020)

On March 12, 2020, ETH fell more than 50% in a few hours. The Flip auction mechanism failed catastrophically: network congestion caused gas prices to spike, preventing most keepers from bidding. Several keepers won auctions with zero-DAI bids, acquiring collateral at zero cost and generating $8.32M in bad debt for the protocol. Cutler (2020) provides mempool-level evidence that the event involved intentional transaction spamming to congest the network.

2.3 The Clipper Mechanism (Liquidations 2.0)

Introduced in 2021, Clipper uses a descending-price Dutch auction. The price starts above market and decreases over time following a linear schedule. The first keeper that calls take() at an acceptable price wins the lot. Flash loans are explicitly supported: keepers can borrow DAI, execute the take, and repay the loan in the same transaction, eliminating the upfront capital requirement. The design was documented in MakerDAO (2022) with the explicit goal of broadening the competitive keeper base.

2.4 The Three Crisis Episodes

This paper analyzes three stress events under both mechanisms, selected for their structural representativeness:

Black Thursday (March 2020). Sudden price shock (-50% in hours) under the Flip mechanism. The liquidation window was 25 hours (category: gradual sustained, per the GSM criterion of §§2.4 applied to the 24h delay in effect at the start of the event). High network congestion, keeper coordination failure, and zero-bid strategy. Represents Flip under extreme stress conditions.

LUNA Crash (May–June 2022). Gradual collapse of the LUNA/UST ecosystem under the Clipper mechanism. Prolonged stress (56 hours of substantive activity distributed over 44 calendar days), high market liquidity, and massive keeper entry. Represents Clipper under moderate, sustained stress conditions.

Tariff Shock (April 2025). On April 2, 2025, the Trump administration announced global tariffs. On April 6, ETH fell ~14% in 24 hours ($1,800 → ~$1,547). A position of 67,570 ETH (~$106M) fell below the 144% collateralization threshold, triggering automatic liquidation. Event concentrated in 67 minutes. Represents Clipper under high-magnitude instantaneous shock conditions.

This paper adopts an operational criterion to classify events by shock speed, anchored in MakerDAO's Governance Security Module (GSM) — whose institutional design and effects on monetary transmission are analyzed in detail in Villega (2026a). The criterion is applied to the reference delay in effect at the start of each event. An event that resolves before the GSM can activate is, by definition, one against which governance has no possible institutional response. On this basis, four categories are defined: instantaneous shock (< 10 hours), rapid response (10–24 hours), gradual sustained (24–48 hours), and gradual extended (> 48 hours). By this criterion, the Tariff Shock (active liquidation window: 4 hours) is instantaneous, Black Thursday (25 hours, with a 24h delay in effect at the start of the event) is gradual sustained, and the LUNA Crash (56 hours) is gradual extended. The Tariff Shock was announced on April 2 and its impact on MakerDAO positions materialized on April 6, but the liquidation of the 67,570 ETH position was concentrated in 67 minutes — a distinction between macroeconomic shock duration and operative liquidation window that the criterion addresses directly. The operational criterion refers to this second dimension. Methodological note: the reference delay for Black Thursday is 24 hours — the value in effect in March 2020, not the current 48 hours (Post-Endgame, May 2025). During the crisis itself, the delay was reduced on an emergency basis to 4 hours (Villega 2026c, §4), which creates an inherent ambiguity in the case: the event defining the category is the same one that altered the classificatory parameter. The criterion is applied to the delay in effect at the start of the shock, before any reactive institutional response; under that criterion (25h > 24h), Black Thursday falls in the gradual sustained category.

2.5 Related Literature

This paper sits at the intersection of four bodies of literature.

Liquidation mechanisms in DeFi. Gudgeon et al. (2020) and Qin et al. (2021) provide the first systematic empirical studies of DeFi liquidation markets. Qin et al. document 2,011 unique liquidator addresses and 28,138 liquidation events. Sadeghi and Feinstein (2026) model optimal liquidation strategy under MEV and oracle manipulation.

Black Thursday and analysis of MakerDAO failure. The zero-bid attack of March 12–13, 2020 is documented in Klages-Mundt and Minca (2020/2021). Cutler (2020) provides mempool-level evidence of intentional transaction spamming.

Dutch auction design. The Clipper mechanism is documented in MakerDAO's technical documentation (2022). Qin et al. (2024) analyze Loss-Versus-Fair efficiency of Dutch auctions on blockchains.

MEV and market concentration. BIS (2022) documents that MEV capture is highly concentrated among a small number of sophisticated actors. Messias and Ferreira Torres (2025) find that control of Arbitrum's Timeboost express lane is highly centralized, with two entities winning more than 90% of auctions.

3. Data

We use raw Ethereum event logs and transaction data from the Dune Analytics platform (DuneSQL). Our primary data sources are: ethereum.logs, ethereum.transactions, ethereum.traces, and erc20_ethereum.evt_transfer.

The Black Thursday sample covers the period March 12–13, 2020 (25-hour window) for event analysis, with the complete Flip period covering January 2020 – July 2021. The LUNA Crash sample covers May–June 2022, extracted from the Clipper contract logs (0xc679...) for that historical period, with 56 hours of substantive activity (threshold ≥ 5 takes/hour). The full Clipper period sample spans January 2023 – March 2026 and is used for aggregate market structure analysis (Section 4) and the Tariff Shock (April 6, 2025, 22:15–23:22 UTC, n = 4 total activity hours; 0xc721 capture window: 67 minutes). The LUNA Crash and full Clipper period sample come from the same contract but different time windows; their data do not overlap. The Clipper sample excludes the 2021–2022 period, during which the volume of ETH-A liquidations via Clipper was insufficient for market structure analysis.

Three complementary metrics are used: (i) the Herfindahl-Hirschman Index (HHI) calculated on auction count and on liquidated collateral volume, (ii) collateral-captured-per-keeper analysis, and (iii) statistical robustness tests using non-parametric bootstrap (10,000 resamples) and the Mann-Whitney test with p-value estimation via Monte Carlo simulation (50,000 permutations).

A note on comparability: Flip uses the Flipper contract (0xd8a0...) under an English auction protocol; LUNA Crash and Tariff Shock use the Clipper contract (0xc679...) under a Dutch auction protocol. The mechanical differences may affect the observed HHI independently of the underlying competitive structure. Results per event are comparable within the same mechanism; comparisons across mechanisms (BT vs. Clipper events) should be interpreted with this caveat.

4. Market Structure Results — Full Period

4.0 HHI Comparability Across Auction Mechanisms

Before reporting the compared HHIs between Flip and Clipper periods, it is necessary to establish the interpretive scope of that comparison. The HHI measures concentration of adjudicated volume, but the formation of that concentration operates in structurally different ways in each mechanism. In Flip (English auction), competition is sequential in time: multiple keepers can participate and bid in windows ranging from minutes to hours, so an entity with high participation arrived at that result by competing against and outbidding others across successive rounds. In Clipper (Dutch auction), competition is for access at a single instant: the first actor to call take() at an acceptable price wins the lot, and the only relevant criterion is execution latency.

While the HHI is heterogeneous across mechanisms in terms of how concentration was formed, it remains a valid measure of ex post value capture in both cases: what matters for concentration analysis is not how concentration was formed but what fraction of value was captured by one actor. The Flip→Clipper comparison should be interpreted as a comparison of capture outcomes, not competitive process. With this explicit caveat, the full-period HHIs are comparable as indicators of the distributive outcome of the mechanism; they are not comparable as indicators of the competitive process that produced them.

4.1 Comparative Statistics: Flip vs. Clipper

Table 1 compares market structure statistics between the full operating periods of both mechanisms. The HHI increase from 1,562 to 2,962 (~90%) coexists with a virtually identical number of active keepers (68 vs. 73), configuring the central speed-concentration trade-off of this paper.

Table 1. Comparative statistics: Flip vs. Clipper (full periods)

Metric	Flip ETH-A (2020–2021)	Clipper ETH-A (2023–2026)
Active keepers	68	73
Total auction events	7,167 tend()*	238 take()
Top-1 share	28.24% (0x6066)	53.78% (0xc721)
Top-2 share	44.84%	~67%
Top-4 share	72.03%	~80%+
HHI (full period)	1,562	2,962
Mechanism	English auction (tend/dent)	Dutch auction (take)

Note: HHI scale 0–10,000. DOJ/FTC thresholds: < 1,500 competitive, 1,500–2,500 moderate, > 2,500 concentrated. *Individual tend() events; the number of unique auctions is lower given each Flip auction may contain multiple tend() calls.

4.2 The Speed-Concentration Design Trade-off

The transition to Clipper produced a comparable number of active keepers (68 vs. 73), but the HHI increased approximately 90% (1,562 → 2,962), indicating substantially greater concentration. The dominant keeper's share nearly doubled (28.24% → 53.78%).

The transition to Clipper produced a comparable number of active keepers (68 vs. 73), but the HHI increased approximately 90% (1,562 → 2,962), indicating substantially greater concentration. The dominant keeper's share nearly doubled (28.24% → 53.78%). the Clipper HHI of 2,962 is calculated over the total cumulative 238 takes of the full period — not as an average of per-event HHIs — and is dominated by the Tariff Shock, which concentrates 128 of those 238 takes (53.78%) in 67 minutes. Excluding the Tariff Shock, the Clipper HHI on the remaining 110 takes (2023–2026 without April 6, 2025) falls well below the DOJ/FTC moderate concentration threshold, consistent with an unconcentrated market. The LUNA Crash, with count HHI ~26, carries low weight in the aggregate HHI precisely because its liquidations are distributed among 399 keepers. The per-event analysis, developed in Sections 5 and 6, is necessary to correctly interpret this aggregate result.

5. Event Analysis: Market Concentration

5.1 HHI per Event: Count and Volume

Table 2 summarizes the weighted HHI for the three crisis episodes. The consistency between count and volume metrics validates the findings independently of the operational choice.

Table 2. Market concentration by event (count and volume HHI)

Event	HHI count	Top-1 count	HHI volume	Top-1 volume	n hours
Black Thursday (Mar-2020)	~1,562	28.24%	~2,100 est.	30.28%	25

LUNA Crash (May–Jun 2022)	~26	0.73%	~221	8.70%	56
Tariff Shock (Apr-2025)	~8,371	91.43%	~9,546	97.68%	4

Note: HHI scale 0–10,000. DOJ/FTC thresholds: < 1,500 competitive, 1,500–2,500 moderate, > 2,500 concentrated. BT volume HHI estimated from top-10 keepers.

5.2 Black Thursday (HHI ~1,562): Moderate Concentration

Black Thursday produced a moderately concentrated market below the DOJ/FTC threshold of 2,500, with the two leading keepers (0x6066 and 0x9c05) capturing 60.4% of volume. The distribution between count and volume is uniform, indicating pre-existing infrastructure advantages rather than execution-speed dominance. The dominant keeper (0x6066, 28.24% of volume) was funded entirely in DAI by a single funder (0x4c8745), which transferred 5,522,979 DAI in 1,229 operations during the crisis window.

Three keepers in the zero-bid cluster (0x9631, 0xb400, 0xb00b) shared a common funder (0xf188761c, ~4.7M DAI combined) and exchanged DAI directly with each other during the event. This pattern is inconsistent with independent competition and indicates coordinated capital infrastructure as the driver of concentration.

5.3 LUNA Crash (Count HHI ~26): Genuine Competition

The LUNA Crash represents the case where Clipper functioned as designed. The mechanism produced massive entry of 399 keepers — 96.74% of them one-timers with no prior activity — with count HHI ~26, close to perfect competition. The elimination of capital barriers via flash loans was effective: the barrier was not capital but the identification of the liquidation opportunity.

However, volume analysis reveals an important asymmetry: the highest-volume keeper captured 8.70% of the ETH liquidated with a single take (2,867 ETH). The LUNA market was not competition among established professional participants, but rather massive opportunistic entry amid accessible liquidations distributed across 56 hours of activity. Competition for access (count HHI ~26) coexisted with moderate asymmetry in value distribution (volume HHI ~221).

Under prolonged stress and deep market conditions, Clipper produces competitive outcomes; the concentration observed in the full Clipper period is largely an artifact of the Tariff Shock.

5.4 Tariff Shock (HHI ~8,371–9,546): De Facto Monopoly

The Tariff Shock produced the most extreme concentration observed in any documented MakerDAO liquidation event. Keeper 0xc721 captured 67,561 ETH (97.68% of the total liquidated; of the original 67,570 ETH position, 9 ETH remained as residual margin) against 1,297 ETH from the second participant — a difference of two orders of magnitude. The 128 takes by 0xc721 were executed within a single 67-minute window (22:15–23:22 UTC on April 6, 2025), competing in real time against 9 additional keepers.

Capital flow analysis reveals that 0xc721 operated without external financing, exclusively via flash loans. The on-chain data from the event reveals its structure: 0xc721 is a single collateral-receiving address served by multiple executing bots with different tx_senders and bot_contracts (19 pairs identified). These 19 pairs are internal components of 0xc721's infrastructure, not independent competitors. Gas profiles of 800k–1.1M per transaction confirm flash loan + swap + take packaged in a single transaction. Concentration is in the collateral destination, not the executor: all bots channel the liquidated ETH to that same address, validating the treatment of 0xc721 as a single actor for HHI calculation purposes. 0xc721's activity is exclusively limited to the Clipper contract 0xc679... during the April 6, 2025 window, with no prior or subsequent records, confirming specialization in high-magnitude instantaneous shocks.

6. Capital Infrastructure Analysis by Keeper

6.1 Top Keepers by Liquidated Volume

Table 3. Top keepers by ETH liquidated

Event / Top Keeper	Takes/Auctions	ETH Liquidated	Share (%)
BT — 0x6066 (1st)	1,056	48,541.9	30.28%
BT — 0x9c05 (2nd)	967	48,220.5	30.08%
LUNA — 0xfdbb (top)	1	2,867.0	8.70%
Tariff — 0xc721 (1st)	128	67,561.0	97.68%
Tariff — 0xb91e (2nd)	4	1,297.0	1.88%

Note: Abbreviated addresses. BT uses Flipper contract (0xd8a0...); LUNA and Tariff use Clipper (0xc679...). Total ETH liquidated in BT: 160,262.4 ETH; BT shares calculated on this total.

6.2 Black Thursday: The Coordinated Funding Cluster

The competition between the two leading keepers in Black Thursday was close in volume (30.28% vs. 30.08%), but funding analysis reveals underlying coordination. 0x6066 received its liquidity from a single centralized funder. Three keepers in the zero-bid cluster shared funder 0xf188761c and exchanged DAI with each other during the event. The gas strategies of the different keepers reveal differentiated archetypes: 0xf1ff paid an average 1,561 gwei (nearly 10x the median) in a gas war strategy, while 0x6066 executed 1,695 total bids — including bids on auctions it did not win — at 161 gwei in a broad-volume strategy, resulting in 1,056 auctions won.

6.3 LUNA Crash: Massive Opportunistic Entry

96.74% of the 399 keepers in the LUNA Crash were one-time participants with no prior activity. The highest-volume keeper captured 8.70% of the ETH with a single large take, but the remaining 91.30% was distributed among 398 keepers. This pattern is consistent with the hypothesis that Clipper democratizes access to moderate-size liquidations, while large positions still tend to be captured by actors better positioned to identify them.

6.4 Tariff Shock: Monopoly by Execution Infrastructure

The asymmetry of the Tariff Shock is striking: 0xc721 liquidated 67,561 ETH while the second participant liquidated 1,297 ETH — a difference of approximately 52x (more than one order of magnitude). The absence of pre-positioned capital and the exclusive operation via flash loans confirm that 0xc721's advantage was not financial but of execution infrastructure: latency, gas optimization, and the ability to package multiple operations within competitive block windows against 9 simultaneous additional keepers.

7. Statistical Robustness Analysis

7.1 Non-Parametric Bootstrap (95% CI)

Non-parametric bootstrap with 10,000 resamples was applied to the hourly HHI distribution of each event, weighting each hour by its number of transactions. Table 4 presents the results.

Table 4. Weighted HHI and 95% bootstrap confidence intervals

Event	Weighted HHI	95% CI Lower	95% CI Upper	n hours
Black Thursday	4,496	3,851	5,522	25
LUNA Crash	1,429	1,124	1,905	56
Tariff Shock	8,868	4,000	~10,000	4

Note: 10,000 resamples, 2.5% and 97.5% percentiles. The BT and LUNA CIs do not overlap; the BT and Tariff CIs partially overlap ([4,000, 5,522]) due to n = 4 for Tariff. LUNA vs. Tariff do not overlap. The weighted HHI in this table (BT: 4,496) differs from the HHI in Table 2 (BT: ~1,562) because they measure different aggregations: Table 2 calculates HHI over the full cumulative event period; Table 4 calculates the transaction-weighted mean of the hourly HHI. Both metrics are correct. With n = 4 hours for the Tariff Shock, the upper CI (~10,000) reflects the ceiling of the HHI index, not additional uncertainty.

The confidence intervals do not overlap between the LUNA Crash and the other two events under any resampling scenario. The partial overlap between Black Thursday and Tariff Shock exclusively reflects the small sample size of Tariff (n = 4 hours), not uncertainty about the direction of the result.

7.2 Mann-Whitney U Tests

The Mann-Whitney U test (non-parametric, two-tailed) was applied to the three possible event pairs. The p-value was estimated by Monte Carlo simulation with 50,000 permutations per pair, an appropriate method given the small sample sizes and non-normality of the hourly HHI distribution.

Table 5. Mann-Whitney U results — pairwise event comparison

Pair Compared	n1	n2	p-value	Effect |r|	Interpretation
BT vs. LUNA Crash	25	56	< 0.001	0.846	Very large — BT more concentrated
LUNA vs. Tariff Shock	56	4	0.005	0.438	Medium — Tariff more concentrated

Note: Monte Carlo, 50,000 permutations per pair. Two-tailed hypothesis. $r = |U - n1 \cdot n2/2| / \sqrt{(n1 \cdot n2 \cdot (n1+n2+1)/12)}$. Large effect threshold: $|r| > 0.5$. The BT vs. Tariff Shock comparison (n1=25, n2=4, p=0.168, |r|=0.314) is omitted from the table due to insufficient statistical power (n_Tariff=4 hours); it is reported here as descriptive reference. The difference in weighted HHI (Tariff: 8,868 vs. BT: 4,496) is robust in volume data and confirmed indirectly by the two adequately-powered pairs.

The tests confirm the ordinal ranking with statistical solidity. Black Thursday is significantly more concentrated than the LUNA Crash ($p < 0.001$, very large effect, $|r| = 0.846$). The Tariff Shock is significantly more concentrated than the LUNA Crash ($p = 0.005$, medium effect, $|r| = 0.438$). The Black Thursday vs. Tariff Shock comparison ($p=0.168$, $|r|=0.314$) is excluded from Table 5 due to insufficient statistical power (n_Tariff=4 hours) and documented in the table note as descriptive reference. The difference in weighted HHI (Tariff: 8,868 vs. BT: 4,496) is reported as descriptive statistics: the Tariff Shock's superiority in concentration is robust in collateral volume data and confirmed indirectly by the two adequately-powered pairs.

The sample size asymmetry (56 vs. 4 for LUNA-Tariff) does not invalidate the test — Mann-Whitney is robust to this condition (Mann and Whitney, 1947) — but reduces statistical power. In this case, the effect is sufficiently large to overcome that limitation.

8. Discussion: Two Paths to Concentration

The three episodes reveal that market concentration in MakerDAO's liquidation mechanisms is not a stable property of mechanism design, but a function of shock speed and intensity. Table 6 synthesizes the two structural paths to concentration documented in this paper.

Table 6. Structural comparison: Flip vs. Clipper

Dimension	Flip ETH-A	Clipper ETH-A
Dominant keeper	0x6066 (28.24%)	0xc721 (53.78% full period; 91.43% Tariff Shock)**
Capital model	Pre-funded DAI via centralized funder	Flash loans — zero upfront capital
Infrastructure type	Coordinated funding network	MEV bot fleet
Concentration driver	Shared capital + zero-bid strategy	Execution speed advantage in stress events
HHI (full period)	1,562	2,962

Note: The comparison aggregates full periods. The Clipper HHI of 2,962 reflects the dominant weight of the Tariff Shock; the baseline HHI of the Clipper mechanism, measured by the LUNA Crash, is ~26 (count) and ~221 (volume). **The dominant Clipper keeper (0xc721) had zero activity outside the Tariff Shock: its full-period share (53.78%) and event share (91.43%) are metrics over different windows.

8.1 The Barrier Shift

Under Flip, the binding constraint was capital. Concentration was driven by funding infrastructure: a coordinated group of keepers shared DAI liquidity networks and executed zero-bid strategies. This barrier was, in principle, addressable through DeFi composability.

Under Clipper, the capital barrier was eliminated by design. Flash loans place all of DeFi's DAI liquidity at the disposal of any keeper. However, concentration increased under fast shocks: the binding constraint shifted from capital to execution infrastructure — latency, contract optimization, gas management, and the ability to compete in real time against multiple bots. This infrastructure cannot be quickly replicated or borrowed.

The pattern is not exclusive to the liquidation layer: Villega (2026c) documents that the dominant voting proxy for the February 2025 governance spell operated with an analogous logic — structured consolidation of 50,000 MKR in a single block 10 days before the cast, coordination of eight wallets in 52 minutes — suggesting that infrastructure asymmetry is a structural property of the protocol as a whole, not an artifact of any specific mechanism (Villega 2026c, §5.2 and §6). The empirical estimation of the κ_coord component across 217 executive spells confirms that coordination cost is a function of capital immobilization and delay, not of voting concentration (Villega 2026d, §5–6; R^2=0.672).

In liquidations, HHI captures the outcome of an instant access process: whoever calls take() first wins the lot, and HHI directly reflects execution infrastructure advantage. In governance, HHI measures the distribution of capital already mobilized at the time of the spell cast — that is, the final result of coordination, not its cost. The coordination effort (organization time, spell rejection risk, capital immobilized before the cast) is not recorded in the spell's HHI: a single actor with 99.7% of MKR can execute a regular spell with near-zero coordination cost, while 17 actors collectively reaching 100% may require substantially greater coordination effort. That is why gsm_delay_hours and mkr_mobilized — the two dimensions of the pre-cast coordination process — dominate HHI as predictors of κ_coord.

The LUNA Crash demonstrates that this barrier shift does not generate inevitable concentration: when the shock is gradual and the market has time to respond, 399 keepers can compete on equal terms. The speed barrier only becomes binding when the liquidation window is sufficiently narrow for differences in execution latency to be the determining factor.

This result is consistent with Lemma 3 of Villega (2026a): the fixed integration cost K — here interpreted as MEV infrastructure cost (latency, contract optimization, gas management) — predicts low n* and endogenous concentration under instantaneous shocks. The analogy is structural: just as Lemma 3 derives that only n*=2 APDs are viable in the Pot given high K, the Tariff Shock reveals that only n*≈1 keeper is viable in a 67-minute window given execution infrastructure asymmetry. The mechanism is the same; the layer, different. The empirical estimation of the κ_coord component — the dominant component of K in the governance layer — is developed in Villega (2026d).

8.2 The Reformulated Trade-off

The speed-concentration trade-off documented in Section 4 requires precision in light of the per-event analysis. The aggregate Clipper period HHI (2,962) is higher than Flip (1,562), but this comparison obscures critical variation: under gradual shocks (LUNA Crash), Clipper produces highly competitive markets (count HHI ~26); under instantaneous shocks (Tariff Shock), it produces extreme concentration (HHI ~8,371–9,546). The trade-off does not reside in the mechanism but in its interaction with shock speed. The aggregate Clipper period HHI is dominated by the Tariff Shock, a 67-minute event in a three-year period.

The policy implication is different from what the aggregate analysis suggests. The problem is not that Clipper is more concentrating than Flip in general — the LUNA Crash suggests it can be more competitive. The problem is that Clipper is extremely concentrating under the precise condition where liquidation is most urgent: the high-magnitude instantaneous shock.

8.3 Policy Implications

These findings suggest that liquidation mechanism design faces a fundamental tension: any mechanism fast enough to protect the protocol from bad debt during instantaneous crashes will also disproportionately reward the actor with the most advanced execution infrastructure. Three design directions emerge from this analysis.

Batch auctions with uniform liquidation price. These would completely eliminate the speed advantage: if all bids within a block or epoch are treated symmetrically, execution latency ceases to be a differentiator. The trade-off is delayed liquidation, which could increase protocol losses during periods of rapid price decline.

Keeper whitelist with capital requirements. This would exchange one form of concentration for another, replacing infrastructure barriers with explicit entry barriers. The cost is reduced decentralization and the governance burden of managing the whitelist.

Protocol-owned liquidation backstop. This would eliminate the keeper market entirely for extreme events, acting as a backstop when concentration exceeds predefined thresholds. This eliminates the winner-take-most dynamic at the cost of introducing risk onto the protocol's balance sheet and requiring reserved capital.

None of these proposals eliminates the fundamental tension. The choice is not between concentration and its absence — it is between different forms of concentration and different risk foci. The LUNA Crash suggests, however, that the tension is manageable under gradual stress conditions: the problem is specific to instantaneous shocks.

9. Conclusion

We document concentration dynamics in MakerDAO's liquidation markets across three crisis episodes and two mechanisms. The results are consistent across metrics, methods, and specifications, and reveal a pattern that aggregate analysis obscures: market concentration is not a stable property of the liquidation mechanism but a function of shock speed.

Black Thursday (Flip, March 2020) produced moderate concentration (HHI ~1,562) driven by coordinated capital infrastructure. The LUNA Crash (Clipper, May–June 2022) produced genuine access competition (count HHI ~26, 399 keepers) with moderate volume asymmetry (volume HHI ~221), demonstrating that Clipper can function as designed under prolonged stress. The Tariff Shock (Clipper, April 2025) produced de facto monopoly (HHI ~8,371–9,546), with a single keeper capturing 97.68% of the liquidated ETH in 67 minutes, driven by MEV execution infrastructure, not capital.

Mann-Whitney tests (Monte Carlo, 50,000 permutations) confirm statistically significant differences in the two pairs with adequate statistical power: BT vs. LUNA ($p < 0.001$, $|r| = 0.846$) and LUNA vs. Tariff ($p = 0.005$, $|r| = 0.438$). The BT vs. Tariff Shock comparison ($n = 4$ hours) is reported as descriptive statistics due to insufficient power (§7.2). The 95% bootstrap intervals do not overlap between LUNA and the other two events.

The central implication for mechanism design is that the Dutch auction solved the right problem — capital barriers — for the wrong type of shock. Under gradual shocks, Clipper produces competitive markets. Under high-magnitude instantaneous shocks, it creates monopoly conditions more extreme than those produced by Flip. The challenge for future design is mechanisms that maintain the openness of the LUNA Crash under the speed of the Tariff Shock — a tension that mechanism design alone may not be completely capable of resolving.

Scope of the study. The results documented in this paper are derived exclusively from the MakerDAO/Sky protocol and the three crisis episodes analyzed. The applicability of the analytical framework — the interaction between shock speed and concentration under Dutch auction mechanisms — to other DeFi protocols with different liquidation logic (Aave, Compound, Uniswap v3) has not been empirically tested. In Aave, for example, liquidations operate with instant price logic without auctions, which alters the nature of the speed-concentration trade-off. Cross-validation in other protocols is a necessary extension before formulating design conclusions with general scope.

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